RAKIA SOUEI

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RESEARCH INTERESTS

Machine learning for medical imaging and clinical decision support; deep learning-based segmentation and anomaly detection; predictive analytics for engineering and environmental systems; natural language processing and large language models; explainable AI (SHAP, LIME); data engineering and ML pipelines.

EDUCATION

B.Eng., Software Engineering – Data Science Track – *Higher Institute of Computer Science* Sept 2023 – Sept 2026
Gabès University, Tunisia

- Grade: 14 / 20 (ranked top 10 in cohort)
- Relevant coursework: Machine Learning, Deep Learning, NLP, Data Visualisation & Analysis, Software Testing, Agile Methodologies.

Preparatory Cycle (Classes Préparatoires en Informatique) – *Higher Institute of Computer Science* Sept 2021 – June 2023
Gabès University, Tunisia

- Mathematics, Statistics, Algorithms, Probability, and Numerical Analysis.

RESEARCH PROJECTS

Research Intern – MITACS Globalink Research Internship – *Concordia University, Dept. of Mechanical & Industrial Engineering* 2026
Montreal, Quebec, Canada

- Implementing a machine learning framework for pipeline leakage prediction in fluid mechanics systems.
- Applying supervised learning models for predictive maintenance and anomaly detection.

Deep Learning Research Project – Final Year project – *Higher Institute of Computer Science* 2025
Gabès University, Tunisia

- Developed an automated 3D MRI spinal cord segmentation pipeline using the U-Net architecture.
- The model achieved 85.53% Dice coefficient and 87.42% IoU on the dataset.
- Built a cross-sectional area (CSA) prediction module to support neurological diagnosis workflows.
- Integrated into a Flutter mobile application for result visualization.
- GitHub Repository | Full Report (PDF)

Soil Moisture Prediction – Geospatial ML – *AI SMART Hackathon (3rd Prize)* 2025
Monastir, Tunisia

- Developed an ensemble ML pipeline to predict soil moisture from Sentinel-2 imagery, SAR radar backscatter, and meteorological time series.
- Combined LightGBM, XGBoost, and CatBoost via weighted ensembling; achieved competitive R^2 on held-out validation data.
- GitHub Repository

TRAINING

Data Analysis Intensive Bootcamp – *GoMyCode* Dec 2025 – Mar 2026
Remote

- Three-month programme: data manipulation, Power BI, end-to-end data extraction, cleaning, transformation, and visualisation.
- Completed a hands-on project applying the full data analysis pipeline to a real-world dataset.

Generative AI with Diffusion Models – *NVIDIA Deep Learning Institute* 2025
Monastir, Tunisia

- Completed training in diffusion model architectures and LLM fine-tuning.

PROFESSIONAL EXPERIENCE

Django Developer Intern – *Squad Crafter* July 2025
Gabès, Tunisia

- Built a Learning Management System (LMS) using Django for a personalized school learning platform.

Software Engineering Intern – *Emertxe India* July – Aug 2024
India (Remote)

- Developed a smart home automation system using Arduino, PICSimLab, and Blynk IoT platform with real-time

mobile monitoring.

TECHNICAL SKILLS

Languages:	Python, SQL, JavaScript, HTML/CSS
ML & DL:	PyTorch, scikit-learn, XGBoost, LightGBM, CatBoost, U-Net, Transformers
Data & Analytics:	NumPy, Pandas, Matplotlib, Seaborn, Power BI, SHAP, LIME (XAI)
Tools & Platforms:	Django, Git & GitHub, Figma, Agile, Agentic Coding (Claude CLI)

AWARDS & HONOURS

International

- IEEE Women in Engineering Distinguished Mentor Award – IEEE WIE (2026).
- Winner – IEEE Computer Society Region 8 "Share Your Journey" Contest, Cairo, Egypt (2024).
- IEEE Region 8 Inspiring Student Ambassador – SIGHT, Humanitarian Technology (2024).

National

- National *Student Entrepreneur* Label – University of Gabès (2026).
- 1st Prize – JPO Innovation Track Idea Competition, University of Gabès (2025).
- 3rd Prize – IEEE AI SMART Hackathon, Monastir, Tunisia (2025).
- 2nd Prize – DEV Hackathon, Medical AI Solution Category (2024).

LEADERSHIP

- **Founder & President** – ARSII (Association de Recherche Scientifique et Innovation en Informatique, ISIMG), 2025–2026.
- **Founder & Chair** – IEEE ISIMG Industry Applications Society (IAS) Student Branch Chapter, 2024–2025.
- **Chair** – Ecoguardians AI Competition 1.0, ISIMG (2024).
- **Volunteer** – IEEE CS International Initiative Team; AIESEC Tunisia (2022–2024); Tunisian Red Crescent (2019–2024).

CONFERENCES & WORKSHOPS

- IEEE SMART AI Summit in Research and Technology, Tunisia (Dec 2025) – *Attendee*.
- IEEE Industry Applications Society Annual Meeting, Sousse, Tunisia (Oct 2025) – *Delegate*.
- IEEE Computer Society Region 8 Congress, Cairo, Egypt (Nov 2024) – *Delegate*.
- IEEE Women in Engineering Annual Congress of Tunisia 4.0, Hammamet (Oct 2024) – *Delegate*.
- Innovation & Entrepreneurship Summer Camp – Avila University / US Embassy Tunisia (2024) – *Selected Participant*.

LANGUAGES

Arabic | French | English